

Section: Our Dynamic Universe

Topic: Measure, Gravitation, Relativity

Question 4 – The DART Asteroid Mission

(a)(i) Compare the mass of Earth and Didymos

✓ 2 marks

Earth : 6.0×10^{24} kg Didymos : 5.3×10^{11} kg

Orders of magnitude difference : $24 - 11 = 13$

✓ **Answer:** The Earth is **13 orders of magnitude more massive** than Didymos.

(a)(ii) Calculate the gravitational force between Earth and Didymos

✓ 3 marks

Use Newton's Law of Gravitation:

$$F = \frac{Gm_1m_2}{r^2}$$

$$G = 6.67 \times 10^{-11}, \quad m_1 = 6.0 \times 10^{24}, \quad m_2 = 5.3 \times 10^{11}, \quad r = 1.1 \times 10^{10}$$

$$F = \frac{6.67 \times 10^{-11} \cdot 6.0 \times 10^{24} \cdot 5.3 \times 10^{11}}{(1.1 \times 10^{10})^2} = 1.75 \times 10^6 \text{ N}$$

(b)(i) Kinetic energy transferred from DART to Dimorphos

✓ 3 marks

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2} \cdot 570 \cdot (6600)^2 = 1.24 \times 10^{10} \text{ J}$$

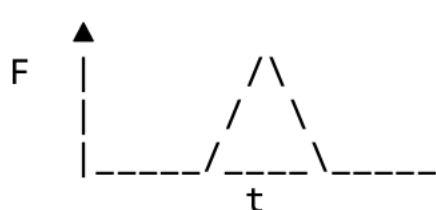
(b)(ii) Force-time graph sketch

✓ 1 mark

A correct sketch shows:

- Force (F) vs Time (t)
- **Smooth, symmetric bell curve**
- Peak force near the middle

✓ **Diagram should look like this:**



Revision Tips

- The **gravitational force** depends on product of masses and inverse square of distance.
- In a **“compare” question**, always use powers of 10 or a clear ratio.
- A **force-time graph** in a collision shows impulse (area under the curve).
- **Kinetic energy** is calculated from $\frac{1}{2}mv^2$ — always check units (m/s!).